Trees

→ Talk about how till now we have discussed on different DS and give example

→ how stack / Queue were actually implemented using the list / LL as well layer above basic → how they make few operation fast.
i.e. Primary vs derived

→ how the initial data type we studied, we didn't went in so much detail about working & complexity and thus having underlying knowledge is important

Tree

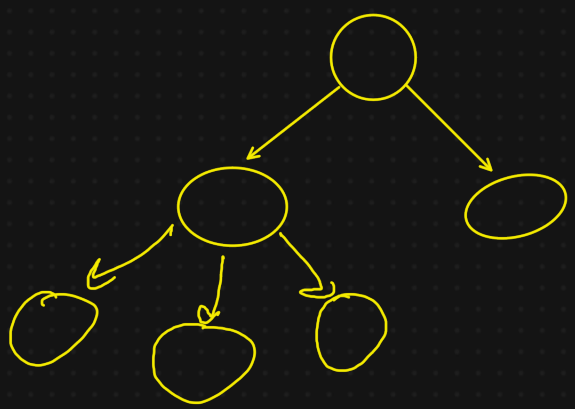
Tree is a heirarchial data structure consisting of nodes connected by edges.

heirarchy

explain how in some use case we might not be easily able to represent data using array / list

eg. family tree
organizational chart
file system

Now make example & explain need & root



→ can be any no. of nodes
→ each subtree should not be joined. (disjoint)

Practical Examples

→ File Directory
→ Decision Tree

Industry use case

→ Indexing in Database
help in fast searches and retrieval

→ Network routing

→ HTML DOM (Document Object model)

Terminologies

1. Node → each item of tree
2. Edge → Connection or link between 2 nodes
3. Root → Top most node in a tree which all other nodes descend from.

4. Leaf → Node with no children
5. Parent
6. Child
7. Sibling
8. Ancestor
9. Descendant

1. Internal Node → neither root nor child
2. Depth / Level → no. of edges from the node to the tree's root . root → 0
3. Height → Longest path from node to a leaf. leaf node → 0
4. Subtree →
5. Path → A sequence of nodes and edges connecting a node to other node.

Implementation of trees

Tree Node
data
children

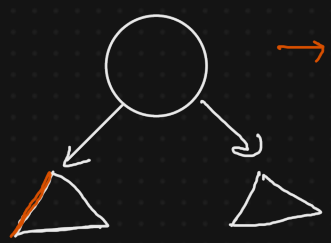
Tree node
data
children



Print a Tree

Take tree input

will get root



→ imagine like this tree has a root & further Sub-tree ⇒ recursion

iterate and run over our tree and also no map etc. as of now.

Generic case

Cannot be sure tree is right

Should rather print nicely

1 : 2, 3,
2 :
3 :

None an issue

Take input → return root of our tree

Shit way to take input
as knowledge of recursion

Taking input using Queue
i.e. level wise
as more intuitive.

Height of tree



how it depends on levels

and also how is it

1 + max ( )

Level / Depth of a node

distance from root

Print all at a particular level

using recursion just level - 1

edge case

very easy

Count leaf node

Print = as of now we have seen Pre order

Next is Post order